

Completion of a Lévy market by power-jump assets

José Manuel Corcuera¹, David Nualart¹, Wim Schoutens²

¹ Universitat de Barcelona, Gran Via de les Corts Catalanes, 585, 08007 Barcelona, Spain
(e-mail: {corcuera,nualart}@mat.ub.es)

² K.U.Leuven, U.C.S., W.De Croylaan 54, 3001 Leuven, Belgium
(e-mail: Wim.Schoutens@wis.kuleuven.ac.be)

Abstract. Except for the geometric Brownian model and the geometric Poissonian model, the general geometric Lévy market models are incomplete models and there are many equivalent martingale measures. In this paper we suggest to enlarge the market by a series of very special assets (power-jump assets) related to the suitably compensated power-jump processes of the underlying Lévy process. By doing this we show that the market can be completed. The very particular choice of the compensators needed to make these processes tradable is delicate. The question in general is related to the moment problem.

Key words: Lévy processes, market models, martingales, SDE, variation, arbitrage, complete markets, moment problem, orthogonal polynomials

JEL Classification: C61

Mathematics Subject Classification (2000): 91B28, 91B26, 91B16, 91B70

1 Introduction

We will work under a so-called (geometric) Lévy market model. Under this model the stock price process $S = \{S_t, t \geq 0\}$ is modelled by a stochastic differential equation (SDE) driven by a general Lévy process $Z = \{Z_t, t \geq 0\}$ (satisfying some conditions):

$$\frac{dS_t}{S_{t-}} = bdt + dZ_t, \quad S_0 > 0. \quad (1)$$

The work of José Manuel Corcuera and David Nualart is partially supported by the MCyT grant no. BFM200304294. W. Schoutens is a Postdoctoral Fellow of the Fund for Scientific Research - Flanders (Belgium) (F.W.O. - Vlaanderen).

Manuscript received: October 2003; final version received: June 2004

If the Lévy process Z is a Brownian motion, this is the classical Black-Scholes model [8], which is *complete*, that is, all contingent claims can be replicated by a self-financing portfolio consisting of investments in the stock and in a risk-free bond. In such a complete model there exists a unique equivalent martingale measure and the unique price of a contingent claim is just the expectation of the discounted payoff at maturity. We will allow more general Lévy processes that take into account jumps. Except for the geometric Brownian model and the geometric Poissonian model, there are many equivalent martingale measures for the above described general geometric Lévy market models and such markets are incomplete: contingent claims cannot in general be hedged by a self-financing portfolio.

In this paper we suggest to enlarge the market by a series of very special assets related to the power-jump processes of the underlying Lévy process. These processes were introduced in [23] and are related with options on the asset, see [2], and with the (realized) variation processes as studied in [4] and [5]. A martingale representation theorem in terms of these (orthonormalized) power-jump processes leads us to market completeness. This completeness can also be used to solve the portfolio optimization by the martingale method, see [14].

For pure jump processes, the power-jump process of order two is just the variation process of degree two, i.e. the quadratic variation process (see e.g. [3, 4] and [5]), and is related with the so-called realized variance. Contracts on realized variance have found their way into OTC markets and are now traded regularly. Higher order power-jump processes have a similar relationship with what one could call realized skewness and realized kurtosis processes. Contracts on these objects however are not common. Carr et al. [11] study contracts on the quadratic variation processes in a model driven by a so-called Sato process. Other references about realized variance products are [10, 13, 16] and [9].

First we recall some basic results on Lévy processes. Next, we introduce the power-jump processes, their compensated versions and the assets by which we will enlarge the market. In a first part we assume that an equivalent martingale measure is given, which we believe is the right one. We then introduce into the market a series of assets based on the above mentioned processes in such a way that the market will remain arbitrage-free. We then show that the market is also complete and we give explicit hedging portfolios for claims whose payoff function only depends on the stock prices at maturity. In a second part we assume the market is already enlarged by these so-called power-jump assets. Here we turn to the question whether the market is arbitrage-free and complete. We examine whether there exists an equivalent martingale measure making all the discounted prices of the traded assets martingales. We thus study the existence of an equivalent martingale measure making not only the discounted stock price a martingale, but also all discounted prices of the newly introduced assets. This question is related to the moment problem. We make the computation explicit in a popular example.

2 The geometric Lévy model

Before describing the model we first recall some basic results on Lévy processes. Basic reference texts on Lévy processes are [6] and [26]. For applications in finance see [27]. We closely follow the presentation given in [12].

2.1 Lévy processes

Given an infinitely divisible distribution with characteristic function $\phi(z)$ one can define a stochastic process $Z = \{Z_t, t \geq 0\}$, called Lévy process, which starts at zero, has independent and stationary increments and such that the distribution of an increment over $[s, s + t]$, $s, t \geq 0$, i.e. $Z_{t+s} - Z_s$, has $(\phi(z))^t$ as characteristic function. It is well-known that Lévy processes are semimartingales.

Such a real-valued stochastic process Z , defined on a complete probability space $(\Omega, \mathcal{F}, P)$, has a càdlàg modification [25, Theorem 30, p. 21] and we will always assume that we are using this càdlàg version. If we let $\mathcal{F}_t = \mathcal{G}_t \vee \mathcal{N}$, where $\mathcal{G}_t = \sigma\{Z_s, 0 \leq s \leq t\}$ is the natural filtration of Z and $\mathcal{N}$ are the P -null sets of $\mathcal{F}$, then $\{\mathcal{F}_t, t \geq 0\}$ is a right-continuous family of σ -fields, [25, Theorem 31, p. 22].

The function $\psi(z) = \log \phi(z) = \log E[\exp(izZ_1)]$ is called the *characteristic exponent* and it satisfies the following *Lévy-Khintchine formula* (see [6]):

$$\psi(z) = i\alpha z - \frac{c^2}{2} z^2 + \int_{-\infty}^{+\infty} (\exp(izx) - 1 - izx1_{\{|x|<1\}}) \nu(dx),$$

where $\alpha \in \mathbb{R}$, $c \geq 0$ and ν is a measure on $\mathbb{R} \setminus \{0\}$ with $\int_{-\infty}^{+\infty} (1 \wedge x^2) \nu(dx) < \infty$. We say that our infinitely divisible distribution has a triplet of Lévy characteristics $[\alpha, c^2, \nu(dx)]$. The measure $\nu(dx)$ is called the *Lévy measure* of Z and dictates how the jumps occur. Jumps of sizes in the set A occur according to a Poisson process with parameter $\int_A \nu(dx)$. If $c^2 = 0$ and $\int_{-1}^{+1} |x| \nu(dx) < \infty$, it follows from standard Lévy process theory (see [6] and [26]) that the process is of finite variation.

From the Lévy-Khintchine formula, one can deduce that Z must be a linear combination of a standard Brownian motion $W = \{W_t, t \geq 0\}$ and a pure jump process $X = \{X_t, t \geq 0\}$:

$$Z_t = cW_t + X_t,$$

and W is independent of X . Moreover,

$$X_t = \int_{\{|x|<1\}} x(Q((0, t], dx) - t\nu(dx)) + \int_{\{|x|\geq 1\}} xQ((0, t], dx) + \alpha t,$$

where $Q(dt, dx)$ is a Poisson random measure on $(0, +\infty) \times \mathbb{R} \setminus \{0\}$ with intensity $dt \times \nu$, where ν is the Lévy measure of Z and dt denotes the Lebesgue measure.

For the purpose of our model we require the process Z to satisfy certain conditions. We will suppose that the Lévy measure satisfies for some $\varepsilon > 0$, and $\lambda > 0$

$$\int_{(-\varepsilon, \varepsilon)^c} \exp(\lambda|x|) \nu(dx) < \infty. \quad (2)$$

This implies that

$$\int_{-\infty}^{+\infty} |x|^i \nu(dx) < \infty, \quad i \geq 1,$$

and that the characteristic function $E[\exp(iuX_t)]$ is analytic in a neighborhood of 0 and

$$E[\exp(-hZ_1)] < \infty \text{ for all } h \in (-h_1, h_2),$$

where $0 < h_1, h_2 \leq \infty$. So, all moments of Z_t (and X_t) exist. Note that

$$\alpha = E[X_1] - \int_{|x| \geq 1} x \nu(dx).$$

With these assumptions one can see that the Doob decomposition of X , in terms of a martingale part and a predictable process of finite variation, is given by

$$X_t = L_t + at,$$

where $L = \{L_t, t \geq 0\}$ is a martingale and $E[X_1] = a$. We refer to the process $t \mapsto at$ as the predictable part or the compensator of X .

If we write $M(dt, dx) = Q(dt, dx) - dt\nu(dx)$ the martingale part of X can be written in terms of the compensated Poisson random measure $M(dt, dx)$ as

$$L_t = \int_{-\infty}^{+\infty} x M((0, t], dx).$$

2.2 Power-jump processes

The following transformations of $Z = \{Z_t, t \geq 0\}$ will play an important role in our analysis. We set

$$Z_t^{(i)} = \sum_{0 < s \leq t} (\Delta Z_s)^i, \quad i \geq 2,$$

where $\Delta Z_s = Z_s - Z_{s-}$, and for convenience we put $Z_t^{(1)} = Z_t$. Note that $Z_t = \sum_{0 < s \leq t} \Delta Z_s$ is not necessarily true; it is only true in the bounded variation case (with c necessarily equal to zero). If we define $X_t^{(i)}$ in an analogous way, we have that $X_t^{(i)} = Z_t^{(i)}$, $i \geq 2$ and clearly $[X, X]_t = X_t^{(2)}$. The processes $X^{(i)} = \{X_t^{(i)}, t \geq 0\}$, $i \geq 2$, are again Lévy processes and are called the i th-power-jump processes (or the power-jump processes of order i). They jump at the

same points as the original Lévy process, but the jumps sizes are the i th power of the jump size of the original Lévy process.

We have $E[X_t] = E[X_t^{(1)}] = ta = tm_1 < \infty$ and (see [25, p. 29])

$$E[X_t^{(i)}] = E\left[\sum_{0 < s \leq t} (\Delta X_s)^i\right] = t \int_{-\infty}^{\infty} x^i \nu(dx) = m_i t < \infty, \quad i \geq 2. \quad (3)$$

We denote by

$$Y_t^{(i)} = Z_t^{(i)} - E[Z_t^{(i)}] = Z_t^{(i)} - m_i t, \quad i \geq 1,$$

the compensated processes. In [23] a procedure is described to orthonormalize the sequence of martingales $\{Y^{(i)}, i \geq 1\}$. By taking a suitable linear combination of the $Y^{(i)}$'s, one obtains a set of pairwise strongly orthonormal martingales $\{T^{(i)}, i \geq 1\}$. Each $T^{(i)}$ is a linear combination of the $Y^{(j)}$, $j = 1, 2, \dots, i$:

$$T^{(i)} = c_{i,i} Y^{(i)} + c_{i,i-1} Y^{(i-1)} + \dots + c_{i,1} Y^{(1)}, \quad i \geq 1.$$

The constants $c_{i,j}$ can be calculated as described in [23]: they correspond to the coefficients of the orthonormalization of the polynomials $\{x^n, n \geq 0\}$ with respect to the measure $\mu(dx) = x^2 \nu(dx) + c^2 \delta_0(dx)$. The resulting processes $T^{(i)} = \{T_t^{(i)}, t \geq 0\}$ are called the *orthonormalized i th-power-jump processes*.

2.3 The geometric Lévy model

Using Itô's formula (see [12] or [25]) for càdlàg semimartingales one can show that (1) has the explicit solution

$$S_t = S_0 \exp\left(cW_t + L_t + \left(a + b - \frac{c^2}{2}\right)t\right) \prod_{0 < s \leq t} (1 + \Delta L_s) \exp(-\Delta L_s).$$

In order to ensure that $S_t \geq 0$ for all $t \geq 0$ almost surely, we need $\Delta L_t \geq -1$ for all t . We thus need that the Lévy measure ν is supported on a subset of $[-1, +\infty)$.

The riskless rate of interest we assume to be a constant r . The value of the riskfree bond or bank account at time t is then given by $B_t = \exp(rt)$.

3 Enlarging the Lévy market model

Suppose we have an equivalent martingale measure Q under which Z remains a Lévy process. Under this measure the discounted stock price process is a martingale and the process $\tilde{Z} = \{Z_t + (b - r)t, t \geq 0\}$ will be a Lévy process (with Lévy measure $\tilde{\nu}$); moreover the process $\tilde{Z}$ is a martingale. Obviously $\Delta \tilde{Z}_t = \Delta Z_t$ and $\tilde{Z}_t^{(i)} = Z_t^{(i)}$, $i \geq 2$. Let us consider (based on $\tilde{Z}$) the i th-power-jump processes $Y^{(i)} = \{Y_t^{(i)}, t \geq 0\}$ and their orthonormalized versions $T^{(i)} = \{T_t^{(i)}, t \geq 0\}$ under the measure Q , i.e. the compensators are $m_i t = tE_Q[\tilde{Z}_1^{(i)}]$

and the orthonormalization procedure is performed under Q . Note that for $i \geq 2$, $m_i = \int_{-\infty}^{+\infty} x^i \tilde{\nu}(dx)$, and we will require $\tilde{\nu}$ to fulfill (2).

We will enlarge the Lévy market with what we will call *i th-power-jump assets*. More precisely, we will allow trade in assets with price processes $H^{(i)} = \{H_t^{(i)}, t \geq 0\}$ where

$$H_t^{(i)} = \exp(rt)Y_t^{(i)}, \quad i \geq 2,$$

and we will denote their orthonormalized versions by $\bar{H}^{(i)} = \{\bar{H}_t^{(i)}, t \geq 0\}$, where

$$\bar{H}_t^{(i)} = \exp(rt)T_t^{(i)}, \quad i \geq 2.$$

Trade in the power-jump assets can be motivated as follows. Consider the 2th-power-jump asset. This object in some sense measures the volatility of the stock since it accounts for the square of the jumps. If one believes that in the future there will be a more volatile environment than the current market's anticipation, trading the 2th-power-jump asset can be of interest. Also if one would like to cover against periods of high (or low) volatility, they can be useful: Buying 2th-power-jump assets can cover the possible losses due to such unfavorable periods. The same can be said for the higher order variation assets. Typically the 3th-power-jump asset is measuring a kind of asymmetry ("skewness") and the 4th-power-jump process is measuring extremal movements ("kurtosis"). Trade in these assets can be of use if one likes to bet on the realized skewness or realized kurtosis of the stock: one believes that the market is not correctly accounting asymmetry and possible extremal moves for. On the other hand, an insurance against a crash can be easily built from the 4th-power-jump (or i th-power jump, $i \geq 4$) assets.

Note that clearly, the discounted versions of the $H^{(i)}$ are the power-jump processes, and hence Q -martingales:

$$E_Q[\exp(-rt)H_t^{(i)}|\mathcal{F}_s] = E_Q[Y_t^{(i)}|\mathcal{F}_s] = Y_s^{(i)}, \quad 0 \leq s \leq t.$$

Hence the market allowing trade in the bond, the stock and the power-jump assets remains arbitrage-free.

3.1 Market completeness

We will make use of the martingale representation property (MRP) obtained in [23] and further exploited in [24], which says that any square-integrable Q -martingale M_t can be represented as follows:

$$M_t = M_0 + \int_0^t h_s d\tilde{Z}_s + \sum_{i=2}^{\infty} \int_0^t h_s^{(i)} dT_s^{(i)},$$

where h_s and $h_s^{(i)}$, $i \geq 2$ are predictable processes such that

$$E\left[\int_0^t |h_s|^2 ds\right] < \infty$$

and

$$E\left[\int_0^t \sum_{i=2}^{\infty} |h_s^{(i)}|^2 ds\right] < \infty.$$

The MRP implies that the market enlarged with the i th-power-jump assets is complete in the sense that for every square-integrable contingent claim X (i.e. a non-negative square-integrable $\mathcal{F}_T$ -measurable random variable) we can set up a sequence of self-financing portfolios whose final values converge in $L^2(Q)$ to X . These portfolios will consist of a finite number of bonds, stocks and i th-power-jump assets. We will say, for short, that X can be replicated. Note that this notion of completeness is equivalent to the notion of *approximate completeness* of Björk et al. given in [7].

Theorem 1 *The Lévy market model enlarged with the i th-power-jump assets is complete, in the sense that any square-integrable contingent claim X can be replicated in the above sense.*

Proof Consider a square-integrable contingent claim X with maturity T . Let

$$M_t = E_Q[\exp(-rT)X|\mathcal{F}_t].$$

By the MRP given above, if we define

$$M_t^N := M_0 + \int_0^t h_s d\tilde{Z}_s + \sum_{i=2}^N \int_0^t h_s^{(i)} dT_s^{(i)},$$

we have that

$$\lim_{N \rightarrow \infty} M_t^N = M_t$$

in $L^2(Q)$. Define the sequence of portfolios (in terms of the orthonormalized i th-power-jump assets)

$$\phi^N = \{\phi_t^N = (\alpha_t^N, \beta_t, \beta_t^{(2)}, \beta_t^{(3)}, \dots, \beta_t^{(N)}), t \geq 0\}, N \geq 2$$

by

$$\begin{aligned} \alpha_t^N &= M_{t-}^N - \beta_t S_{t-} e^{-rt} - e^{-rt} \sum_{i=2}^N \beta_t^{(i)} \bar{H}_{t-}^{(i)}, \\ \beta_t &= e^{rt} h_t S_{t-}^{-1}, \\ \beta_t^{(i)} &= h_t^{(i)}, \quad i = 2, 3, \dots, N. \end{aligned}$$

Here α_t^N corresponds to the number of bonds at time t ; β_t is the number of stocks at that time and $\beta_t^{(i)}$ is the number of assets $\bar{H}^{(i)}$, $i = 2, 3, \dots, N$, that one needs

to hold at time t . We claim that $\{\phi^N, N \geq 2\}$ is the sequence of self-financing portfolios which replicates X . In fact, the value V_t^N of ϕ^N at time t is given by

$$V_t^N = \alpha_t^N e^{rt} + \beta_t S_t + \sum_{i=2}^N \beta_t^{(i)} \bar{H}_t^{(i)} = e^{rt} M_t^N,$$

so the sequence of portfolios $\{\phi^N, N \geq 2\}$ is replicating the claim. Denote by

$$G_u^N = r \int_0^u \alpha_t^N e^{rt} dt + \int_0^u \beta_t dS_t + \sum_{i=2}^N \int_0^u \beta_t^{(i)} d\bar{H}_t^{(i)}$$

the gain process, i.e. the gains or losses obtained up to time u by following ϕ^N . We will show that

$$G_u^N + M_0 = M_u^N e^{ru}$$

which implies that the portfolio is self-financing. We have

$$\begin{aligned} G_u^N &= r \int_0^u M_{t-}^N e^{rt} dt - r \int_0^u h_t e^{rt} dt - r \sum_{i=2}^N \int_0^u h_t^{(i)} \bar{H}_{t-}^{(i)} dt \\ &\quad + \int_0^u h_t e^{rt} S_{t-}^{-1} dS_t + \sum_{i=2}^N \int_0^u h_t^{(i)} d\bar{H}_t^{(i)}. \end{aligned} \quad (4)$$

Now, integrating by parts we can write

$$\begin{aligned} r \int_0^u M_t^N e^{rt} dt &= e^{ru} M_u^N - M_0 \\ &\quad - \int_0^u h_t e^{rt} d\tilde{Z}_t - \sum_{i=2}^N \int_0^u h_t^{(i)} e^{rt} dT_t^{(i)}, \end{aligned} \quad (5)$$

and substituting (5) into (4) yields

$$\begin{aligned} G_u^N &= e^{ru} M_u^N - M_0 - r \int_0^u h_t e^{rt} dt - r \sum_{i=2}^N \int_0^u h_t^{(i)} \bar{H}_{t-}^{(i)} dt \\ &\quad - \int_0^u h_t e^{rt} d\tilde{Z}_t + \int_0^u h_t e^{rt} S_{t-}^{-1} dS_t \\ &\quad + \sum_{i=2}^N \int_0^u h_t^{(i)} d\bar{H}_t^{(i)} - \sum_{i=2}^N \int_0^u h_t^{(i)} e^{rt} dT_t^{(i)} \\ &= e^{ru} M_u^N - M_0 - r \int_0^u h_t e^{rt} dt \\ &\quad - \int_0^u h_t e^{rt} d\tilde{Z}_t + \int_0^u h_t e^{rt} S_{t-}^{-1} dS_t \\ &= e^{ru} M_u^N - M_0. \end{aligned}$$

□

3.2 Hedging portfolios

In this section we show how to compute explicitly the sequence of portfolios that replicates a contingent claim whose payoff is only a function of the value at maturity of the stock price, i.e. $X = f(S_T)$.

We start with noting that the value of the contingent claim at time t is given by

$$F(t, S_t) = \exp(-r(T-t))E_Q[f(S_T)|\mathcal{F}_t];$$

we call $F(t, x)$ the price function of X .

Denote by D_1 the differential operation with respect to the first variable, i.e. the time variable, and by D_2 the differential operator with respect to the space variable (the second variable, the stock price). Finally, denote by $\mathcal{D}$ the following integral operator:

$$\mathcal{D}F(t, x) = \int_{-\infty}^{+\infty} (F(t, x(1+y)) - F(t, x) - xyD_2F(t, x)) \tilde{\nu}(dy).$$

It is well-known, see [12], that in analogy with the Black-Scholes partial differential equation, in the Lévy market setting F will satisfy a partial differential integral equation (PDIE). More precisely, the price function $F(t, x)$ satisfies

$$D_1F(t, x) + rxD_2F(t, x) + \frac{1}{2}c^2x^2D_2^2F(t, x) + \mathcal{D}F(t, x) = rF(t, x) \quad (6)$$

with $F(T, S_T) = f(S_T)$.

We will need the following lemma:

Lemma 2 *Consider a real function $h(s, x, y)$ on $\mathbb{R}^+ \times \mathbb{R}^+ \times \mathbb{R}$ which is infinitely differentiable in the y -variable and satisfies $h(s, x, 0) = 0$ and $\frac{\partial h}{\partial y}(s, x, 0) = 0$. Set*

$$a_i(s, x) = \frac{1}{i!} \frac{\partial^i h}{\partial y^i}(s, x, 0),$$

and assume that

$$\sup_{x < K, s \leq s_0} \sum_{i=2}^{\infty} |a_i(s, x)| R^i < \infty, \quad (7)$$

for all $K, R > 0, s_0 > 0$. Then we have

$$\begin{aligned} \sum_{t < s \leq T} h(s, S_{s-}, \Delta X_s) &= \sum_{i=2}^{\infty} \int_t^T \frac{1}{i!} \frac{\partial^i}{\partial y^i} h(s, S_{s-}, 0) dY_s^{(i)} \\ &\quad + \int_t^T \int_{-\infty}^{\infty} h(s, S_{s-}, y) \tilde{\nu}(dy) ds \end{aligned}$$

Proof The function $h(s, x, y)$ can be expanded as

$$h(s, x, y) = \sum_{i=2}^{\infty} a_i(s, x) y^i.$$

Then we have

$$\begin{aligned} \sum_{t < s \leq T} h(s, S_{s-}, \Delta X_s) &= \sum_{t < s \leq T} \sum_{i=2}^{\infty} a_i(s, S_{s-}) (\Delta X_s)^i \\ &= \sum_{i=2}^{\infty} \int_t^T a_i(s, S_{s-}) dX_s^{(i)} \\ &= \sum_{i=2}^{\infty} \int_t^T a_i(s, S_{s-}) dY_s^{(i)} + \sum_{i=2}^{\infty} \int_t^T a_i(s, S_{s-}) m_i ds, \end{aligned}$$

where the sums converge for every $\omega \in \Omega$ because of (7). Using that $m_i = \int_{-\infty}^{\infty} y^i \tilde{\nu}(dy)$, we have

$$\begin{aligned} \sum_{t < s \leq T} h(s, S_{s-}, \Delta X_s) &= \sum_{i=2}^{\infty} \int_t^T a_i(s, S_{s-}) dY_s^{(i)} \\ &\quad + \int_t^T \int_{-\infty}^{\infty} \sum_{i=2}^{\infty} a_i(s, S_{s-}) y^i \tilde{\nu}(dy) ds \\ &= \sum_{i=2}^{\infty} \int_t^T a_i(s, S_{s-}) dY_s^{(i)} \\ &\quad + \int_t^T \int_{-\infty}^{\infty} h(s, S_{s-}, y) \tilde{\nu}(dy) ds. \quad \square \end{aligned}$$

Now we calculate the sequence of self-financing portfolios that replicates the contingent claim X .

Theorem 3 *The sequence of self-financing portfolios replicating a contingent claim X with a payoff only depending on the stock price value at maturity and a price function $F(t, x) \in C^{1,\infty}$ which satisfies*

$$\sup_{x < K, t \leq t_0} \sum_{n=2}^{\infty} |D_2^n F(t, x)| R^n < \infty \quad (8)$$

for all $K, R > 0, t_0 > 0$, is given at time t by

$$\begin{aligned} \text{number of bonds} &= \alpha_t^N = B_t^{-1} (F(t, S_{t-}) - S_{t-} D_2 F(t, S_{t-})) \\ &\quad - B_t^{-1} \sum_{i=2}^N \frac{S_{t-}^i D_2^i F(t, S_{t-})}{i! B_t} H_{t-}^{(i)}, \\ \text{number of stocks} &= \beta_t = D_2 F(t, S_{t-}), \end{aligned}$$

$$\text{number of } i\text{th-power-jump assets} = \beta_t^{(i)} = \frac{S_{t-}^i D_2^i F(t, S_{t-})}{i! B_t}, \quad i = 2, 3, \dots, N.$$

Proof First, apply Itô's lemma to $F(T, S_T)$. This gives

$$\begin{aligned} F(T, S_T) - F(t, S_t) &= \int_t^T D_1 F(s, S_{s-}) ds + \frac{c^2}{2} \int_t^T S_{s-}^2 D_2^2 F(s, S_{s-}) ds \\ &\quad + \int_t^T D_2 F(s, S_{s-}) dS_s \\ &\quad + \sum_{t < s \leq T} F(s, S_s) - F(s, S_{s-}) - D_2 F(s, S_{s-}) \Delta S_s. \end{aligned}$$

Noting that $\Delta S_s = S_{s-} \Delta X_s$ this becomes

$$\begin{aligned} &F(T, S_T) - F(t, S_t) \\ &= \int_t^T D_1 F(s, S_{s-}) ds + \frac{c^2}{2} \int_t^T S_{s-}^2 D_2^2 F(s, S_{s-}) ds \\ &\quad + \int_t^T D_2 F(s, S_{s-}) dS_s \\ &\quad + \sum_{t < s \leq T} F(s, S_{s-}(1 + \Delta X_s)) - F(s, S_{s-}) - D_2 F(s, S_{s-}) \Delta X_s S_{s-}. \end{aligned}$$

Let $h(t, x, y) = F(t, x(1 + y)) - F(t, x) - xy D_2 F(t, x)$. Then

$$\begin{aligned} h(t, x, 0) &= 0, \\ \frac{\partial}{\partial y} h(t, x, 0) &= 0, \\ \frac{\partial^n}{\partial y^n} h(t, x, 0) &= x^n D_2^n F(t, x), \quad n \geq 2. \end{aligned}$$

Since $F(t, x) \in C^{1,\infty}$ satisfies condition (8), h satisfies the conditions in Lemma 2. Applying Lemma 2 gives

$$\begin{aligned} &F(T, S_T) - F(t, S_t) \\ &= \int_t^T D_1 F(s, S_{s-}) ds + \frac{c^2}{2} \int_t^T S_{s-}^2 D_2^2 F(s, S_{s-}) ds \\ &\quad + \int_t^T D_2 F(s, S_{s-}) dS_s + \sum_{i=2}^{\infty} \int_t^T \frac{S_{s-}^i D_2^i F(s, S_{s-})}{i!} dY_s^{(i)} \\ &\quad + \int_t^T \int_{-\infty}^{\infty} (F(t, x(1 + y)) - F(t, x) - xy D_2 F(t, x)) \tilde{\nu}(dy) ds \\ &= \int_t^T \frac{D_1 F(s, S_{s-}) + S_{s-}^2 \frac{c^2}{2} D_2^2 F(s, S_{s-}) + \mathcal{D}F(t, x)}{r B_s} dB_s \\ &\quad + \int_t^T D_2 F(s, S_{s-}) dS_s + \sum_{i=2}^{\infty} \int_t^T \frac{S_{s-}^i D_2^i F(s, S_{s-})}{i! B_s} dH_s^{(i)} \\ &\quad - \sum_{i=2}^{\infty} \int_t^T \frac{S_{s-}^i D_2^i F(s, S_{s-})}{i! B_s} Y_{s-}^{(i)} dB_s. \end{aligned}$$

Finally, using the PDIE (6) for the price, we obtain

$$\begin{aligned}
 & F(T, S_T) - F(t, S_t) \\
 &= \int_t^T \frac{F(s, S_{s-}) - S_{s-} D_2 F(s, S_{s-}) - \sum_{i=2}^{\infty} \frac{S_{s-}^i D_2^i F(s, S_{s-})}{i! B_s} H_{s-}^{(i)}}{B_s} dB_s \\
 &+ \int_t^T D_2 F(s, S_{s-}) dS_s + \sum_{i=2}^{\infty} \int_t^T \frac{S_{s-}^i D_2^i F(s, S_{s-})}{i! B_s} dH_{s-}^{(i)}. \quad (9)
 \end{aligned}$$

□

Remark 4 In the Black-Scholes model the risk-neutral dynamics of the stock price is given by the stochastic differential equation

$$\frac{dS_t}{S_t} = (r - \frac{1}{2}\sigma^2)dt + dW_t, \quad S_0 > 0,$$

where $W = \{W_t, t \geq 0\}$ is a standard Brownian motion. In this case, all processes $H^{(i)}, i \geq 1$ are equal to zero. Hence, it is clear the market is already complete and that an enlargement is not necessary. Moreover the hedging portfolio is given by $\frac{F(s, S_s) - S_s D_2 F(s, S_s)}{B_s}$ (number of bonds) and $D_2 F(s, S_s)$ (number of stocks).

Remark 5 In the Poisson case, the risk-neutral dynamics of the stock price is given by

$$\frac{dS_t}{S_{t-}} = (r - \lambda)dt + dN_t, \quad S_0 > 0,$$

where $N = \{N_t, t \geq 0\}$ is a Poisson process with intensity parameter $\lambda > 0$. Note that in this case

$$S_t = S_0 \exp((r - \lambda)t) 2^{N_t},$$

and that all compensated power-jump processes equal the compensated Poisson process:

$$Y_t^{(i)} = N_t - \lambda t, \quad i \geq 2.$$

Consider a contingent claim with a payoff $F(T, S_T)$ only depending on the value of the stock price at maturity T . Note that in this very special case the payoff is thus also a function of the Poisson process at time T . We make this clear using the notation: $G(T, N_T) = F(T, S_T)$. Note that $G(T, N_T + 1) = F(T, 2S_T)$. Using the relation

$$\begin{aligned}
 & \sum_{i=2}^{\infty} \frac{S_{s-}^i D_2^i F(s, S_{s-})}{i!} \\
 &= F(s, 2S_{s-}) - F(s, S_{s-}) - S_{s-} D_2 F(s, S_{s-}) \\
 &= G(s, N_{s-} + 1) - G(s, N_{s-}) - S_{s-} D_2 F(s, S_{s-}),
 \end{aligned}$$

the equality (9) becomes

$$\begin{aligned} & F(T, S_T) - F(t, S_t) \\ &= \int_t^T \frac{F(s, S_{s-}) - G(s, N_{s-} + 1) + G(s, N_{s-})}{B_s} dB_s \\ &\quad + \int_t^T (G(s, N_{s-} + 1) - G(s, N_{s-})) S_{s-}^{-1} dS_s. \end{aligned}$$

This formula can also be directly derived using Itô's lemma for the Poisson process and the function G . In conclusion, it is clear the market is already complete and that an enlargement is not necessary. Moreover the hedging portfolio is given by $\frac{2G(s, S_{s-}) - G(s, N_{s-} + 1)}{B_s}$ (number of bonds) and $(G(s, N_{s-} + 1) - G(s, N_{s-})) S_{s-}^{-1}$ (number of stocks).

3.3 Pricing formulas

Consider the value at time t of a contingent claim X with a payoff function $f(S_T)$ only depending on the stock price at maturity:

$$F(t, S_t) = \exp(-r(T-t)) E_Q[X | \mathcal{F}_t] = \exp(-r(T-t)) E_Q[f(S_T) | \mathcal{F}_t].$$

Note, because we are working risk-neutrally, that $a + b = r$. Then

$$\begin{aligned} & F(t, x) = \exp(-r(T-t)) \times \\ & E_Q \left[f \left(x e^{c(W_T - W_t) + (L_T - L_t) + \left(r - \frac{c^2}{2}\right)(T-t)} \prod_{t < s \leq T} (1 + \Delta L_s) e^{-\Delta L_s} \right) \right] \\ &= \exp(-r(T-t)) \times \\ & E_Q \left[f \left(x e^{cW_{T-t} + L_{T-t} + \left(r - \frac{c^2}{2}\right)(T-t)} \prod_{0 < s \leq T-t} (1 + \Delta L_s) e^{-\Delta L_s} \right) \right]. \end{aligned}$$

If we introduce the function

$$F_{BS}(t, x) = \exp(-r(T-t)) E_Q \left[f \left(x \exp \left(cW_{T-t} + \left(r - \frac{c^2}{2}\right)(T-t) \right) \right) \right],$$

which gives the price of the option for the Black-Scholes model with volatility c , we have

$$F(t, x) = E_Q \left[F_{BS} \left(t, x e^{L_{T-t}} \prod_{0 < s \leq T-t} (1 + \Delta L_s) e^{-\Delta L_s} \right) \right].$$

Note that the n th derivative with respect to x , needed in the formula for the number of n th power-jump assets in the replicating portfolio, is just

$$D_2^n F(t, x) = E_Q \left[e^{nL_{T-t}} \prod_{0 < s \leq T-t} (1 + \Delta L_s)^n e^{-n\Delta L_s} \right. \\ \left. \times D_2^n F_{BS} \left(t, x e^{L_{T-t}} \prod_{0 < s \leq T-t} (1 + \Delta L_s) e^{-\Delta L_s} \right) \right].$$

Recall that in the case of a classical European option, the $D_2^n F_{BS}$ are very simple. In case of the European call for example, the first two derivatives are given in terms of the cumulative probability distribution function $N(x)$ and the density function $n(x)$ of a standard normal random variable by

$$D_2^1 F_{BS}(t, x) = N(d_1) = N \left(\frac{\log(x/(T-t)) + (r + c^2/2)(T-t)}{c\sqrt{T-t}} \right)$$

and

$$D_2^2 F_{BS}(t, x) = n(d_1)/(xc\sqrt{T-t}),$$

which are also known as the *delta* and the *gamma* of the option and where

$$d_1 = \frac{\log(x/(T-t)) + (r + c^2/2)(T-t)}{c\sqrt{T-t}}.$$

4 Arbitrage

We assume our market is already enlarged with the power-jump assets. So, we have chosen constants $a^{(i)}$, $i \geq 2$ and trade is allowed in the bond, the stock and the power-jump assets with price processes $H_t^{(i)} = \exp(rt)(X_t^{(i)} - a^{(i)}t)$, $i \geq 2$.

We investigate whether this enlargement leads to arbitrage or not. For instance, if we choose $a^{(i)}$ and r equal zero this leads to arbitrage opportunities because all $H_t^{(i)}$ with even i are strictly increasing and starting at zero and trade is allowed in these objects. Actually, the choice of the constants $a^{(i)}$ may prevent arbitrage opportunities. We will discuss below how to make this choice, which is a delicate matter.

No arbitrage, in the usual sense and in our portfolios with a finite number of assets, is implied by the existence of an equivalent martingale measure under which all discounted assets in the market are martingales. This existence question is related with the moment problem and we will give sufficient conditions to ensure that there exists an equivalent martingale measure and hence the market is arbitrage free. But in continuous time, the existence of an equivalent martingale measure is a sufficient but not a necessary condition to ensure no-arbitrage, see [15]. The problem in its full generality seems to be very hard and challenging.

4.1 Equivalent martingale measures

In this section we will describe the measures, equivalent to the canonical (real world) measure under which the discounted stock price process is a martingale and under which Z remains a Lévy process. More precisely, we characterize all *structure preserving* P -equivalent martingale measures Q under which Z remains a Lévy process and the process $\tilde{S} = \{\tilde{S}_t = \exp(-rt)S_t, t \geq 0\}$ is an $\{\mathcal{F}_t\}$ -martingale. Since we are considering a market with finite horizon T , local equivalence will be the same as equivalence.

We have the following result (see [26, Theorem 33.1]):

Theorem 6 *Let Z be a Lévy process with Lévy triplet $[\alpha, c^2, \nu(dx)]$ under some probability measure P . Then the following two conditions are equivalent.*

- (a) *There is a probability measure Q locally equivalent to P for any $t \geq 0$ such that Z is a Q -Lévy process with triplet $[\tilde{\alpha}, \tilde{c}^2, \tilde{\nu}(dx)]$.*
- (b) *All of the following conditions hold.*
 - (i) $\tilde{\nu}(dx) = H(x)\nu(dx)$ for some Borel function $H : \mathbb{R} \rightarrow (0, \infty)$.
 - (ii) $\tilde{\alpha} = \alpha + \int_{-\infty}^{+\infty} x \mathbf{1}_{\{|x| \leq 1\}} (H(x) - 1) \nu(dx) + Gc$ for some $G \in \mathbb{R}$.
 - (iii) $\tilde{c} = c$.
 - (iv) $\int_{-\infty}^{+\infty} (1 - \sqrt{H(x)})^2 \nu(dx) < \infty$.

The equivalent conditions in the previous theorem imply that the process $\tilde{W} = \{\tilde{W}_t, t \geq 0\}$ with

$$\tilde{W}_t = W_t - Gt$$

is a Brownian motion under Q and also, if ν and $\tilde{\nu}$ verify the condition (2), the process X is a quadratic pure jump Lévy process with Doob-Meyer decomposition

$$X_t = \tilde{L}_t + (a + \int_{-\infty}^{+\infty} x(H(x) - 1)\nu(dx))t,$$

where $\tilde{L} = \{\tilde{L}_t, t \geq 0\}$ is a Q -martingale and the Lévy measure is given by $\tilde{\nu}(dx) = H(x)\nu(dx)$.

We now want to find an equivalent measure Q under which the discounted price process $\tilde{S}$ is a martingale. By the above theorem, under such a Q , X has the Doob-Meyer decomposition

$$X_t = \tilde{L}_t + (a + \int_{-\infty}^{+\infty} x(H(x) - 1)\nu(dx))t,$$

where $\tilde{L} = \{\tilde{L}_t, t \geq 0\}$ is a Q -martingale. Noting that $\Delta L_t = \Delta \tilde{L}_t$, we have

$$\begin{aligned} \tilde{S}_t &= S_0 \exp \left(c\tilde{W}_t + \tilde{L}_t + \left(a + b - r + cG - \frac{c^2}{2} \right) t \right) \\ &\quad \times \exp \left(t \int_{-\infty}^{+\infty} x(H(x) - 1)\nu(dx) \right) \prod_{0 < s \leq t} (1 + \Delta \tilde{L}_s) \exp(-\Delta \tilde{L}_s). \end{aligned}$$

Then a necessary and sufficient condition for $\tilde{S}$ to be a Q -martingale is the existence of G and $H(x)$ with $\int_{-\infty}^{\infty} (1 - \sqrt{H(x)})^2 \nu(dx) < \infty$ such that

$$cG + a + b - r + \int_{-\infty}^{+\infty} x(H(x) - 1) \nu(dx) = 0. \quad (10)$$

Remark 7 We remark (see e.g. [17]), that if there exists a (non-structure preserving) equivalent martingale measure Q_1 under which Z is not a Lévy process, there always exists a (structure preserving) equivalent martingale measure Q_2 under which Z is a Lévy process.

A sufficient condition to guarantee that the enlarged market is free of arbitrage is the existence of an equivalent martingale measure Q making $\tilde{S}$ and the discounted $H^{(i)}$'s martingales. If this measure is structure preserving, the condition that the discounted stock price must be a martingale comes down to the existence of G and $H(x)$ such that (10) holds. If we want also that the discounted $H^{(i)}$'s, i.e. $X_t^{(i)} - a^{(i)}t$ be martingales for $i \geq 2$, using (3) together with the fact that the Lévy measure of X under Q is given by $H(x)\nu(dx)$ shows that this comes down to

$$\int_{-\infty}^{+\infty} x^i H(x) \nu(dx) = a^{(i)}, \quad i \geq 2. \quad (11)$$

The question now is, whether there exist G and $H(x)$ such that (10) and (11) hold simultaneously. This is related with the moment problem: given a series of numbers $\{\mu_n\}$, find necessary and sufficient conditions for the existence of such measure with μ_n as n th moment. Another point is the uniqueness. A partial result is that if the moment problem has a solution with bounded support, then it will be unique, see [28] or [1]. We then have the following proposition:

Proposition 8 *Suppose that $\nu(dx)$ has compact support. Then if there is a martingale measure in the market enlarged with the power-jump assets, the martingale measure is unique, structure preserving and the market is complete.*

Proof If we have a martingale measure in the enlarged market, there exists, using the same arguments as in [17], an $H(x)$ verifying (10) and (11) with $H(x) > 0$. The measure $\mu(dx) = x^2 H(x) \nu(dx)$ is finite and has a bounded support. This implies that $H(x)$ is determined by condition (11). On the other hand, since the support is bounded, $H(x)\nu(dx)$ verifies (2) and the model enlarged with the power-jump assets is complete. Finally, since the contingent claim $B_T \mathbf{1}_A$ with $A \in \mathcal{F}$ is replicable, the uniqueness of its initial arbitrage price $E_Q(\mathbf{1}_A)$ implies the uniqueness of the martingale measure. $\square$

In general, uniqueness of the martingale measure implies completeness.

Proposition 9 *If the probability measure that makes the discounted stock price and the power-jump assets martingales is unique, the market is complete.*

Proof Let Q be a martingale measure. If the market is not complete, there exists a contingent claim $X \geq 0$, $X \in L^2(Q)$, not identically zero, which is orthogonal to any replicable contingent claim. Define $Q^*(d\omega) = (1 + X)Q(d\omega)$. Then Q^* is a martingale measure different from Q . In fact, for any $s \leq t$, and $A \in \mathcal{F}_s$, we have

$$E_{Q^*}(\mathbf{1}_A(Y_t^{(i)} - Y_s^{(i)})) = E_Q(\mathbf{1}_A(Y_t^{(i)} - Y_s^{(i)})) + E_Q(X\mathbf{1}_A(Y_t^{(i)} - Y_s^{(i)})) = 0,$$

and $\{Y_t^{(i)}, t \geq 0\}$ are Q^* -martingales for all $i \geq 2$. Clearly $\tilde{S}$ is also a Q^* -martingale. $\square$

4.2 Example: a Brownian motion plus a finite number of Poisson processes

Suppose

$$Z_t = cW_t + \sum_{j=1}^n c_j N_{j,t},$$

where $c \neq 0$, $W = \{W_t, t \geq 0\}$ is a standard Brownian motion and $N_j = \{N_{j,t}, t \geq 0\}$ are independent Poisson processes with intensity $a_j > 0$. The constants $c_j, j = 1, \dots, n$ are assumed to be all different from each other and non-zero. Then $X_t = \sum_{j=1}^n c_j N_{j,t}$, $E[X_1] = \sum_{j=1}^n c_j a_j = a$ and

$$H_t^{(i)} = \exp(rt) \left(\sum_{j=1}^n c_j^i N_{j,t} - a^{(i)} t \right), \quad i = 2, 3, \dots$$

It is not that hard to see that $H_t^{(i)}$ for $i > n+1$ can be written as a linear combination of the $H_t^{(i)}, i = 2, \dots, n+1$ (see [20]). In this case we enlarge the market with only n objects, namely the assets following the price processes $H_t^{(i)}, i = 2, \dots, n+1$. In order that an equivalent martingale measure Q exists, we must have the existence of G and H such that

$$\begin{aligned} \int_{-\infty}^{+\infty} x(H(x) - 1)\nu(dx) &= r - cG - a - b \\ \int_{-\infty}^{+\infty} x^i H(x)\nu(dx) &= a^{(i)}, \quad i = 2, \dots, n+1. \end{aligned}$$

The support of H will now be the set $\{c_2, \dots, c_{n+1}\}$ and the above equations reduce to

$$\begin{aligned} \sum_{j=1}^n c_j H(c_j) a_j &= r - cG - b \\ \sum_{j=1}^n c_j^i H(c_j) a_j &= a^{(i)}, \quad i = 2, \dots, n+1. \end{aligned}$$

There exists an equivalent martingale measure if the following system of equations for $H(c_j)$, $j = 1, \dots, n$ has a positive solution, i.e. $H(c_j) > 0$, $j = 1, \dots, n$:

$$\begin{bmatrix} c_1^2 a_1 & c_2^2 a_2 & \dots & c_n^2 a_n \\ c_1^3 a_1 & c_2^3 a_2 & \dots & c_n^3 a_n \\ \dots & \dots & \dots & \dots \\ c_1^{n+1} a_1 & c_2^{n+1} a_2 & \dots & c_n^{n+1} a_n \end{bmatrix} \times \begin{bmatrix} H(c_1) \\ H(c_2) \\ \dots \\ H(c_n) \end{bmatrix} = \begin{bmatrix} a^{(2)} \\ a^{(3)} \\ \dots \\ a^{(n+1)} \end{bmatrix}.$$

The existence (and uniqueness by Proposition 8) of a positive solution $H(c_j)$, $j = 1, \dots, n$ can be translated into the condition

$$C^{-1}a' > 0, \quad (12)$$

where C^{-1} is the inverse of the Vandermonde matrix

$$C = \begin{bmatrix} 1 & 1 & \dots & 1 \\ c_1 & c_2 & \dots & c_n \\ \dots & \dots & \dots & \dots \\ c_1^{n-1} & c_2^{n-1} & \dots & c_n^{n-1} \end{bmatrix}$$

and a' is the transpose of $[a^{(2)} \dots a^{(n+1)}]$. Note that if all c_i 's are different from each other (as we assumed above) then, $\det C \neq 0$.

For the calculation of the inverse of Vandermonde matrices see [18] or [21]. For other applications of Vandermonde matrices in finance see [22].

Remark 10 In case $n = 2$, assuming $c_2 > c_1$, condition (12) can be written as $a^{(2)}c_2 > a^{(3)} > a^{(2)}c_1$.

References

1. Ahiezer, N.I.: The classical moment problem. New York: Hafner Publ. Co. 1965
2. Balland, P.: Deterministic implied volatility models. *Quantitative Finance* **2**, 31–44 (2002)
3. Barndorff-Nielsen, O.E., Shephard, N.: Estimating quadratic variation using realised variance. *J. Appl. Econ.* **17**, 457–477 (2002)
4. Barndorff-Nielsen, O.E., Shephard, N.: Realised power variation and stochastic volatility models. *Bernoulli* **9**, 243–265 (2003)
5. Barndorff-Nielsen, O.E., Shephard, N.: Financial volatility: Stochastic volatility and Lévy based models. Cambridge: Cambridge University Press 2004 (forthcoming)
6. Bertoin, J.: Lévy processes. (Cambridge Tracts in Mathematics 121) Cambridge: Cambridge University Press 1996
7. Björk, T., Di Masi, G., Kabanov, Y., Runggaldier, W.: Towards a general theory of bond markets. *Finance Stochast.* **1**, 141–174 (1997)
8. Black, F., Scholes, M.: The pricing of options and corporate liabilities. *J. Polit. Econ.* **81**, 637–654 (1973)
9. Carr, P., Lewis, K.: Corridor variance swaps. *Risk* **17**, 67–72 (2002)
10. Carr, P., Madan, D.: Towards a theory of volatility trading. In: Jarrow, R. (ed.) *Volatility, risk publications* 1998, pp. 417–427. Reprinted in: Musiela et al. (eds.) *Option pricing, interest rates, and risk management*. Cambridge: Cambridge University Press 2001, pp. 458–476
11. Carr, P., Geman, H., Madan, D.H., Yor, M.: Pricing options on realized variance. *Prépublications du Laboratoire de Probabilités et Modèles Aléatoires* **768**, Universités de Paris 6 & Paris 7, Paris 2002

12. Chan, T.: Pricing contingent claims on stocks driven by Lévy processes. *Ann. Appl. Probab.* **9**, 504–528 (1999)
13. Chriss, N., Moroko, W.: Market risk for volatility and variance swaps. *Risk*, July (1999)
14. Corcuera, J.M., Guerra J., Nualart, D., Schoutens, W.: Optimal investment in a Lévy market. Preprint IMUB 349, Universitat de Barcelona (2004)
15. Delbaen, F., Schachermayer, W.: A general version of the fundamental theorem of asset pricing. *Math. Ann.* **300**, 463–520 (1994)
16. Derman, E., Zou, J., Demeterfi, K., Kamal, M.: A guide to variance swaps. *J. Derivatives* **6**(4), 9–32 (1999)
17. Eberlein, E., Jacod, J.: On the range of option prices. *Finance Stochast.* **1**, 131–140 (1997)
18. Graybill, A.: *Matrices with applications to statistics* (2nd ed.) Belmont: Wadsworth 1983
19. Jacod, J., Shiryaev, A.N.: *Limit theorems for stochastic processes*. Berlin Heidelberg New York: Springer 1987
20. Léon, J.A., Vives, J., Utzet, F., Solé, J.L.: On Lévy processes, Malliavin calculus and market models with jumps. *Finance Stochast.* **6**, 197–225 (2002)
21. Macon, N., Spitzbart, A.: Inverses of Vandermonde matrices. *Am. Math. Monthly* **65**, 95–100 (1958)
22. Norberg, R.: On the Vandermonde matrix and its role in mathematical finance. Laboratory of Actuarial Mathematics (working paper), University of Copenhagen (1999)
23. Nualart, D., Schoutens W.: Chaotic and predictable representations for Lévy processes. *Stochast. Proc. Appl.* **90**, 109–122 (2000)
24. Nualart, D., Schoutens W.: Backwards stochastic differential equations and Feynman-Kac formula for Lévy processes, with applications in finance. *Bernoulli* **7**, 761–776 (2001)
25. Protter, Ph.: *Stochastic integration and differential equations*. Berlin Heidelberg New York: Springer 1990
26. Sato, K.: *Lévy processes and infinitely divisible distributions*. Cambridge Studies in Advanced Mathematics 68. Cambridge: Cambridge University Press 2000
27. Schoutens, W.: *Lévy processes in finance: Pricing financial derivatives*. Chichester: Wiley 2003
28. Shohat, J.A., Tamarkin, J.D.: *The problem of moments*. New York: Mathematical Surveys No. 1, A.M.S. 1950

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.